

UNIVERSITY OF LONDON

MSc Data Science
DSM100 Artificial Intelligence

Coursework assignment
Development and implementation of a Bayesian network

This coursework assignment is weighted at 50% of the final mark for the module.

The purpose of this coursework is to help you to gain a deeper understanding of decision support systems (DSS), their purpose, design approaches, and implementation, to:

- reflect on your research interests in artificial intelligence.
- gain practical experience in conceptual design of a DSS.
- gain practical experience in developing and implementing a DSS.

Your main objective for this assignment is to explore how Bayesian networks can be used in decision support systems.

DSS is an information system (or sub-system) that enhances the decision-making abilities – by a person, software agent, robot, or a group of agents. DSS requires the specification of tasks, objectives, and constraints. Various techniques and approaches can be used for the design and implementation of DSS. One of the most popular techniques for the design and implementation of DSS is probabilistic reasoning, and this is what we are focusing on in this module.

For this assignment, you will choose a specific problem domain (e.g., medical diagnosis, environmental risk assessment, supply chain management) and build a Bayesian network that can be used for decision-making in this context. You will then perform inference using the network, and demonstrate how it assists in making informed decisions.

© University of London

Your tasks are:

Task 1. Conceptual design of a DSS

a. Explain the purpose of your DSS.
b. Provide specification for DSS tasks, objectives, and constraints.
c. Describe data and knowledge that are required for the design, implementation and functioning of your DSS. Explain if it can improve its performance over time and how it can do so.
d. Outline potential stakeholders and users, and how they can benefit from your DSS.
e. Indicate if it can be a stand-alone system, or if it can be integrated with other systems; explain how.

Task 2. Development of a simple Bayesian Network

a. Provide a textual description of what is known.
b. Identify 4–6 nodes representing real-world concepts (e.g., medical diagnosis, weather prediction, or student performance) and links between them.
c. Represent the background knowledge as a Bayesian network. Define the corresponding conditional probability tables (CPTs) manually or based on available data.

Bonus: You will receive extra points if your network consists of more than 6 nodes.

Task 3. Implementation of the Bayesian network in Python

a. Encode the Bayesian network in Python utilising available libraries.
b. Formulate three questions for your DSS and use the implemented Bayesian network to answers them (e.g., “What is the probability of node X given node Y?”).

Bonus Task. Causal inference with Bayesian networks

a. Build a Bayesian network model for a simple system (e.g., smoking causes lung cancer, using variables like smoking habits, family history, pollution, and lung cancer). Or extend accordingly the Network built for Tasks 1-3.
b. Use the network to estimate the causal effect of different variables on a selected node, e.g. how smoking and pollution causing effect on lung cancer.
c. Perform interventions (e.g., set variables to specific values) and assess the changes in the probabilities of outcomes.

You can follow the example in topic 4 - DSS that can advise on the best way to contact your boss – by email or skype, depending on the likelihood of the boss being in office. Such an advice can be given based on the background knowledge. We discussed that the considered DSS can be integrated with other systems, e.g., Alexa.

Note: **A brief description** of the key elements of the considered DSS is in the Appendix. Your description is expected to be more elaborate and detailed than the provided example.

© University of London

Format requirements

Your coursework report should be word- or TeX (LaTeX)-processed. Please submit your report as a pdf or word document, and figures as jpeg images. The formatting requirements are:

- a clear font (Arial, Calibri, Times New Roman)
- 12-point size font
- at least 2cm margins
- number the pages of your text in the bottom right corner of each page
- do not use endnotes or footnotes
- the title page, figures, tables (excluding their captions), appendices do not count towards the word limit.

Please submit your code as an appendix to your report – all in one document.

Number of words limit

Please do not exceed the word count limit of 2,000 words (excluding the title, figures, tables, references, and the code). If you submit more than 2,000 words, the following penalties apply:

1. You should not exceed the maximum word limit. 5 marks will be deducted if the word count is up to 10% more than the maximum word limit.
2. If the word count exceeds the maximum word limit by more than 10%, you will receive a mark of zero for your work.

Please do not treat the limit of 2,000 words as a target. However, we do not recommend submitting reports with fewer than 1,000 words.

Assessment

Specification

Provide a brief description of your DSS and specification of tasks, objectives, and constraints for your DSS.

[10 points]

Conceptual design

Describe the overall architecture of your DSS: What are the components? Is it a stand-alone system? Can it be integrated with other systems? Specify input data and knowledge; how can they be obtained? What are the outputs? What are potential risks in the development of such a system? Who are the potential stakeholders and users of you DSS?

[10 points]

© University of London

Representation of background knowledge as a Bayesian network

Provide textual description of what is known. Explain how you selected nodes and links for the construction of your Bayesian network. This Bayesian network is the core of your DSS. The recommendation (but not the requirement) is to limit the number of nodes by 4-6. How did you determine the associated probabilities?

[15 points]

Bonus

You will receive extra points if your network consists of more than 6 nodes. The network must be coherent and correspond to the specified background knowledge.

[10 points]

Implementation of the Bayesian network

Implement your Bayesian network in Python; include informative comments to your code. Submit your code as an appendix of your course work report, preferably in a pdf format.

[15 points]

Querying Bayesian network

Formulate three questions (queries) for your DSS and use the implemented Bayesian network to answers them. You will receive up to 5 points for each query. The more interesting question is, the more points you will get.

[15 points]

Bonus: Causal inference with Bayesian networks.

Use the network to estimate the causal effect of different variables on a selected node, e.g. how smoking and pollution causing effect on lung cancer.

[5 points]

Perform interventions (e.g., set variables to specific values) and assess the changes in the probabilities of outcomes.

[10 points]

Total number of words

Insert a word count.

The content within the main body of text comprises the overall word count, including in-text citations, references, quotes, heading and sub-headings. The cover page, figures, tables, reference list and any appendices do not count towards the overall word count. **The word limit is 2,000 words.**

[END OF COURSEWORK ASSIGNMENT]

© University of London

Appendix: a brief description of the example DSS

Note: this is **a brief description** of the key elements of the considered in topic 4 system. Your coursework report is expected to be more elaborate and detailed than the text below.

1. Conceptual design

For the considered example, specifications can be formulated as follows:

- **Task:** advise on the best way to contact your boss
- **Objectives:** (1) provide a conceptual description of enhancing Alexa by additional functionality of choosing the way of contacting your boss; (2) represent the background knowledge as a Bayesian network; (3) implement and test the Bayesian network in Python.
- **Constraints**: limit the number of nodes in the Bayesian network by 4-6.

Potential users of the example DSS are office workers. The system can be easily modified for other scenarios, for example how to contact your friend, or neighbour, etc. In that case, the system can be of use to a wide range of users.

It can be **a stand-alone system**, or it can be integrated with other systems, for example with a calendar of my boss. Calendar data will provide knowledge when the boss is away on a business trip, or busy otherwise. Taking into account such additional knowledge will improve the accuracy of estimates of likelihood of a response and consequently quality of advice.

Data and knowledge: the example DSS requires a background knowledge model that specifies types of interactions with the boss and estimates of likelihoods of a response depending on various situations. Such estimates can be obtained based on the data about the previous communications with the boss. For functioning, the system requires a command as an input: contact my boss.

The system **can improve its performance over time** by collecting more data about communications with the boss and making more accurate estimates of likelihoods of a response depending on various situations. The system can be further improved by its integration with other systems, e.g. a calendar, Siri, Cortana.

2. Bayesian network

Textual description of the background knowledge:

I can contact my boss by email or skype. I can get a response or not. If she is in office (only 50% of time as she is often away on business trips), she will respond in 50% cases. The boss usually does not check skype messages when in office but checks 50% of the time when not in office. The boss usually checks email messages when in office and is unlikely to check it when not in office. The boss is very likely to respond, if she has checked both skype and e-mail. The boss is likely to respond, if has checked skype or email. The boss does not respond, if has not checked either skype or email.

© University of London

Selection of the nodes and links: selected an ordered list of random variables, where causes precede effects. Links represent the connection between the nodes.

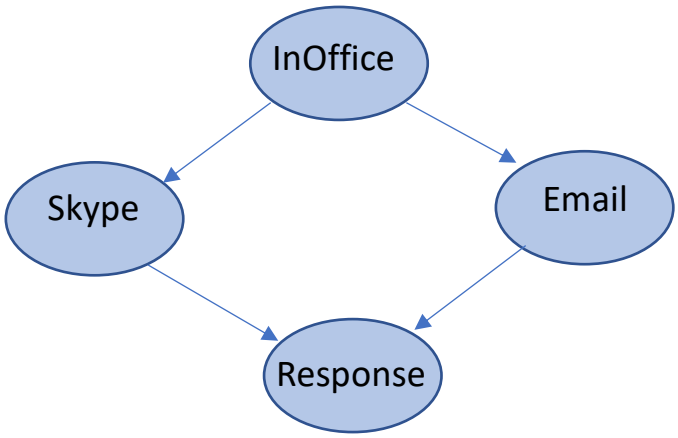
Associated probabilities: Some probabilities are estimated based on the previous experience, e.g. 'The boss is in office in 50% of cases':

P(boss is in office) = 0.5

Reasonable numerical estimates were made to represent uncertain knowledge, e.g. 'The boss usually does not check skype messages when in office'. We can assign an arbitrary probability 0.1 to represent the uncertain expression 'usually does not check':

P(checks skype| boss is in office) = 0.1

Graphical representation:



In topic 4, we considered the implementation of this system as a Bayesian network. Please refer to the provided implementation for further particulars.

[END OF APPENDIX]

© University of London